

Sample End Semester Question Paper - 2025

Subject: Network Theory | Code: EC202 | Semester: 2

Branch: ECE | Regulation: R20 (sample) | Time: 3 Hours | Max Marks:

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. Name two two-port parameters.
2. What is duality in networks?
3. Define natural response.
4. What is an ideal transformer?
5. State one network theorem.
6. State Kirchhoff's current law.
7. What is a planar network?
8. Define Thevenin voltage.
9. What is a first-order circuit?
10. State the maximum power transfer condition.

PART B - Long questions (6 x 13 = 78 Marks)

1. Solve a given circuit using nodal and mesh analysis.
2. Explain Thevenin's, Norton's, superposition and maximum power transfer theorems with examples.
3. Explain transient analysis of first-order RL and RC circuits.
4. Explain two-port network parameters (Z, Y, ABCD, h) and interconnections.
5. Explain series and parallel resonance with examples.
6. Explain dependent sources and duality in network analysis.